NIKHIL BHAGAT

AI/ML Engineer & NLP Researcher | LLM Agents, Generative AI, RAG & Information Retrieval, AI Safety & Evaluation

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PROFESSIONAL SUMMARY

AI/ML engineer and independent natural language processing (NLP) researcher with 6+ years building production backends, machine learning (ML) and generative AI systems for clients in Hong Kong, Japan, Italy, the Philippines and Nepal. Lead senior AI engineer (contract) at a Kathmandu AI company, building 24/7 autonomous, tool-using agents on WhatsApp, Instagram and web chat for real estate, restaurant, travel and accounting businesses. Research focus: retrieval-augmented generation (RAG), evaluation benchmarks and guardrails that keep large language models (LLMs) truthful in low-resource languages; author of NEPVERSA (under review, ACM TALLIP). I build systems that have to be right, not just fluent.

EDUCATION

Tribhuvan University, Mahendra Morang Adarsh Multiple Campus, Biratnagar, Nepal **2025**
Bachelor of Computer Application (BCA), four-year program

RESEARCH

Independent Researcher, Nepali Legal Information Retrieval | Mentor: Prof. Achut Raj Kattel, Tribhuvan University **May 2026 – Present**

- Built **NEPVERSA**, the first version-aware Nepali statutory retrieval benchmark: 5 Acts, 455 provisions, 99 amendment operations and 181 point-in-time and supersession questions (61.3% abstention-expected), with gold answers derived deterministically from amendment footnotes instead of an LLM judge.
- Recovered amendment history where Gazette notices are scanned images (50 of 53 had no text layer) and official PDFs extract to corrupted Unicode; found 7 silent extraction defects and pinned each with mutation-tested regression tests (219 tests) and an automated cross-signal validator.
- Shipped one-command offline reproduction that re-verifies every published number and all 455 provisions by SHA-256; code MIT, annotations CC BY 4.0: github.com/NikeGunn/nyasathi-reserch

PUBLICATIONS

- **Bhagat, N.** (2026). NEPVERSA: A Version-Aware Nepali Statutory Retrieval Benchmark Derived from Amendment Footnotes. Preprint, Zenodo. doi.org/10.5281/zenodo.22916911. Under review at ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP); arXiv (cs.CL, cs.IR) submitted.

PROFESSIONAL EXPERIENCE

Lead Senior AI Engineer (Contract), Kritrim Baudhikata Anusandhan Kendra Nepal Pvt. Ltd. (Kribaat), Kathmandu **Oct 2024 – Present**

- **Kribaat** (kribaat.com): built 24/7 autonomous customer-service agents for real estate agencies and restaurants across WhatsApp, Instagram and web chat in Nepali, Romanized Nepali, English and Chinese. Tool-calling loop (understand, look up, verify, reply) with a verification gate that checks every price, listing ID and booking code against live tool results before sending, plus one-tap human handoff.
- **HisabKitab** (hisabkitab.pro): WhatsApp-native AI accountant for VAT-registered SMEs. Extracts bills from photos and PDFs, applies Nepal IRD VAT (13%) and TDS rules in exact integer paisa, saves only after owner confirmation, and verifies Khalti payments server-side. TypeScript microservices with Model Context Protocol (MCP) tool servers for ledger, payments and Tally; about 1,600 automated tests in CI.
- Resolved a production multilingual-drift incident (Romanized-Nepali users receiving English replies; agent overstating its capabilities) with durable per-user reply-language state, a three-locale message catalog, a meaning-preserving rewrite guard for figures, dates and negation, and capability statements grounded in configuration.
- **Rehearsal Lab** (github.com/NikeGunn/hisabkitab, agents-learning): offline agent evaluation and reinforcement learning environment. The agent replays seeded synthetic bookkeeping scenarios (12 families x 10 instances, train/dev/test splits) scored by a deterministic judge against a hidden oracle; GRPO policy training, LangSmith tracing, crash-safe Postgres workers with hash-chained audit logs, and a release gate where every candidate must beat production before human approval. Database access isolated to a rehearsal-only schema.

- **NyayaSathi** (nyaya.kribaat.com): offline legal research assistant for Nepali advocates over a 337-document law collection (330 Supreme Court judgments and 7 statutes; 11,587 passages). Five-stage verification (citation resolves, exact quote, in force on the as-of date, entailment); 0 fabricated citations on an adversarial set; Precision@10 0.871, Recall@5 0.968, MRR 0.925; p90 17.2 s on an 8 GB laptop. Runs extractive, local Ollama (Qwen2.5-7B) or cloud LLM modes.
- **MIG Alpha** (migalpha.com), Hong Kong travel and visa agency (client): built the Django REST backend and admin, an AI travel-assistant service and an omnichannel social inbox (Facebook Messenger), with Dockerized deployment, monitoring and 169 automated tests.

Senior Backend Engineer (Contract), CineArtLink, Hong Kong (remote) **Jun 2022 – Present**

- Lead backend development for CineArtLink (cineartlink.com), a professional network connecting film, media and theatre talent with projects, built on Django REST Framework.
- Designed and deployed Docker microservices on Heroku and lead migration planning to AWS; built engagement-based feed ranking, CI/CD with GitHub Actions, JWT authentication and Redis caching; mentor junior developers on test-driven development.

Independent Contract Engineer, clients in Japan, Hong Kong, Italy, the Philippines and Nepal **Feb 2020 – Dec 2025**

- **Tung Tung** (Japan): real-time backend for a GPS multiplayer game; PostGIS geofencing and spatial queries, Celery and Redis workers, Firebase push notifications, live top-100 leaderboard.
- **Wahi AI** (Nepal): bilingual English/Nepali RAG assistant for Nepal Financial Reporting Standards (FAISS, LangChain, Django REST Framework).
- **Battisputali** (Hong Kong): backend for a short-video community app with 10k+ users; fixed upload timeouts with chunked streaming to Cloudinary.
- **TaskPilot** (Philippines): eliminated recurring WebSocket drops in a real-time Kanban board with Socket.IO heartbeats.
- **DevLinker** (Italy): full-stack developer network in React and Express; cut build time by 40%.

PROJECTS AND OPEN SOURCE

- **AgentGuard**, zero-trust security sidecar for AI agents (Go, MIT) | agentguard.space | github.com/NikeGunn/agentguard: transparent proxy firewall for MCP and A2A tool calls in Claude Code, Cursor, Codex, Gemini CLI and Windsurf. Five-stage inspection pipeline against prompt injection, tool poisoning, schema rug pulls, secret leakage and runaway loops; YAML policy engine, server trust scores, session replay; under 5 ms p99 overhead, 250+ tests across 6 platforms, cosign-signed releases.
- **PalikaBook**, civic anti-corruption evidence platform for Nepal's 753 local governments (Go, Flutter, Next.js): anonymous, tamper-proof video evidence with chunk hashing and Merkle-root verification, court-ready chain of custody, and a government triage portal with OTP + TOTP login, encrypted messaging and audit logs. PostgreSQL, Redis, MinIO, NATS JetStream, RS256 JWT, AES-256-GCM. Private repository; demo on request.
- **SANU**, Nepali-English LLM fine-tune (independent research) | huggingface.co/Haubaa/SANU-AI-7B-v0.1-GGUF: QLoRA (4-bit NF4, rank 16) fine-tune of Qwen2.5-7B-Instruct on a 290-sample proof-of-concept set including Romanized Nepali and function-calling examples; released GGUF (Q4_K_M, Q8_0) for local Ollama and llama.cpp inference on zero-cost compute. Its failures on Nepali legal and civic questions motivated the NEPVERSA benchmark.
- **AutoGPT**, merged pull request #11618 (github.com/Significant-Gravitas/AutoGPT/pull/11618): replaced a multiprocessing Manager queue with a thread-safe queue in ExecutionQueue, removing per-execution subprocess spawning and IPC overhead.
- **Code language model**: decoder-only transformer written from scratch in PyTorch, trained on 50K+ Python/JavaScript/TypeScript files, with int8 quantization and FastAPI serving. Also built Yeti Agent (browser agent), Hauba (coding agent on LiteLLM), TuTu Engine (offline local AI runtime) and Clawdboost (npm, 1k+ weekly downloads).

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, SQL, Dart

AI / ML: machine learning, deep learning, generative AI, large language models (LLMs), LLM agents, agentic workflows, tool calling, Model Context Protocol (MCP), retrieval-augmented generation (RAG), information retrieval (BM25, dense retrieval, embeddings, semantic search), vector databases (FAISS, Pinecone), fine-tuning (QLoRA, LoRA, PEFT), reinforcement learning (GRPO), LLM evaluation, guardrails, prompt engineering, quantization (int8, GGUF), PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, LangSmith, Ollama, llama.cpp, Claude API, OpenAI API

NLP and Research: natural language processing (NLP), multilingual and low-resource NLP, legal NLP, benchmark and dataset construction, evaluation design, AI safety, reproducible research pipelines

Backend and Data: Django, Django REST Framework, FastAPI, Node.js, PostgreSQL, PostGIS, Redis, MongoDB, Kafka, NATS, Celery, MinIO, REST APIs, WebSockets, microservices, distributed systems, system design

Cloud and DevOps: Docker, Kubernetes, AWS, Terraform, GitHub Actions, CI/CD, Linux | **Frontend and Mobile:** React, Next.js, Flutter, Tailwind CSS